Data Sciences Institute at the University of Toronto

<u>Machine Learning Software Foundations Certificates</u> **- Cohort 8 (07.03.2026)**

Disease Prediction using Machine Learning

Logistic Regression vs Random Forest

Project Contribution Write-Up ML -6

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Introduction

•This group project focused on building a machine learning pipeline to predict diseases based on patient-reported symptoms.

•The dataset was obtained from Kaggle and contained 132 binary symptom featur and one target variable representing 41 different diseases.

•The objective was to explore how machine learning models can identify patterns between symptoms and diseases and support early disease prediction.

•My contributions included both technical development and team coordination.

My project workflow

Data Exploration & Preparation

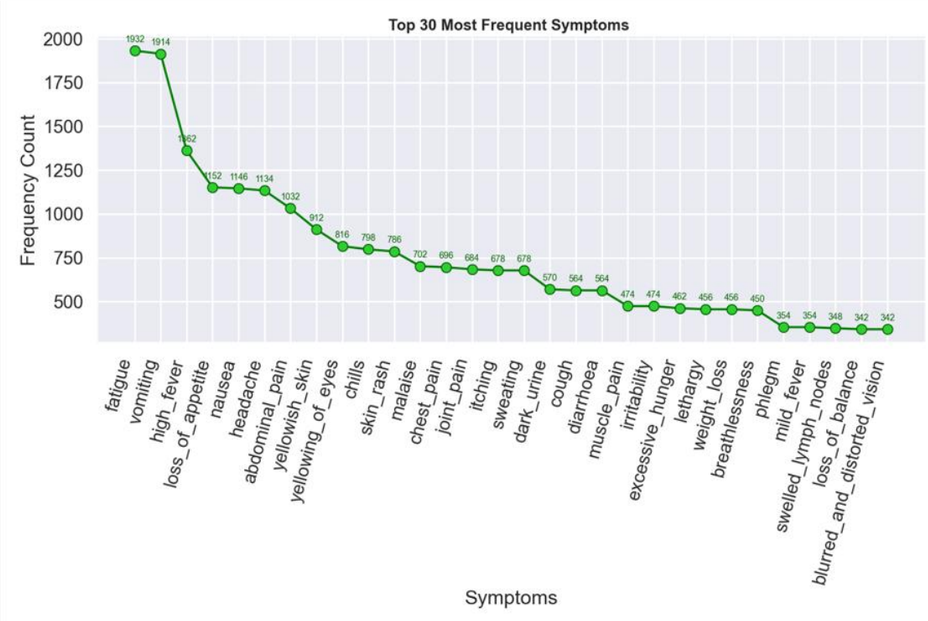
Performed **Exploratory Data Analysis (EDA)**.

- I performed **Exploratory Data Analysis** to better understand the dataset.

- I analyzed the **frequency of symptoms**, the **distribution of diseases**, and the **number of symptoms per patient record**.

- On average, each patient record contained about **seven symptoms**. I also created visualizations to understand which symptoms occur most frequently.

- Since machine learning models require numeric input, I used **label encoding** to convert disease names into numeric values.



Model Development

- On the technical side, I worked on implementing and evaluating classification models for disease prediction. Specifically, I developed and tested Logistic Regression and Random Forest models for the multi-class classification task.

- I applied stratified 5-fold cross-validation to improve model reliability and used GridSearch hyperparameter tuning to identify the optimal parameters for model performance.

Logistic Regression

The first model is **Multinomial Logistic Regression**, which is suitable for multi-class classification problems.

Trained the model using the training dataset and used it to predict diseases in the test dataset.

Logistic regression works by estimating the probability that a symptom combination belongs to a specific disease class.

Random Forest

The second model is **Random Forest**, an ensemble learning algorithm.

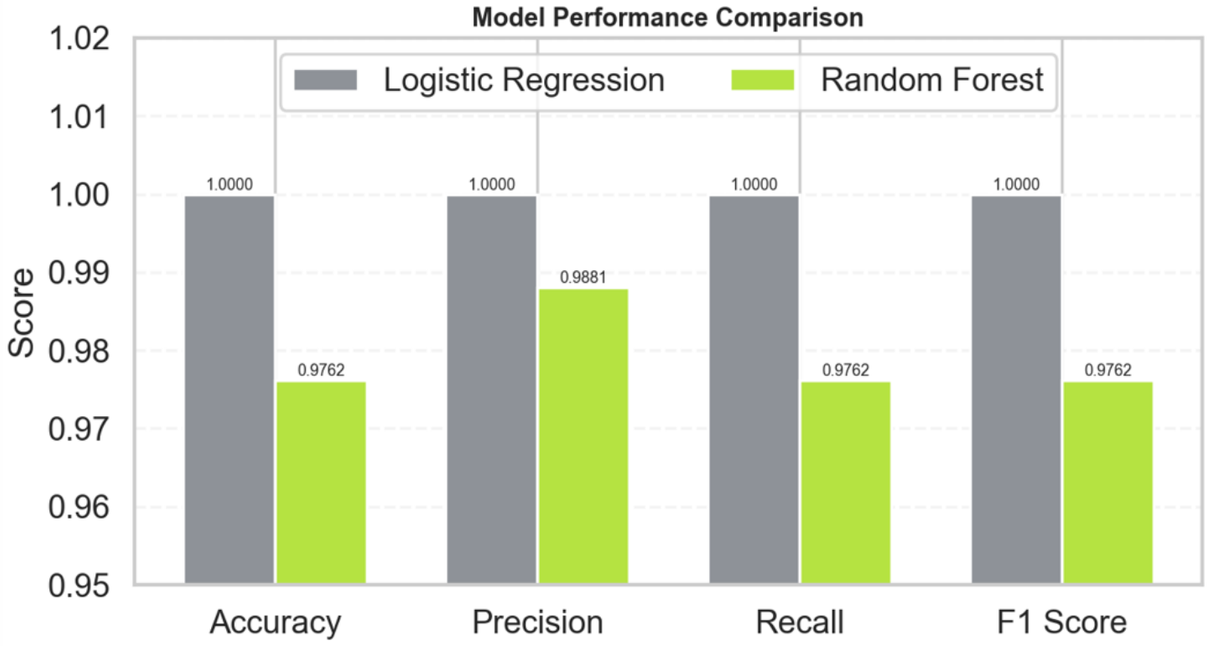
Random Forest works by building multiple decision trees and combining their predictions.
Applied:

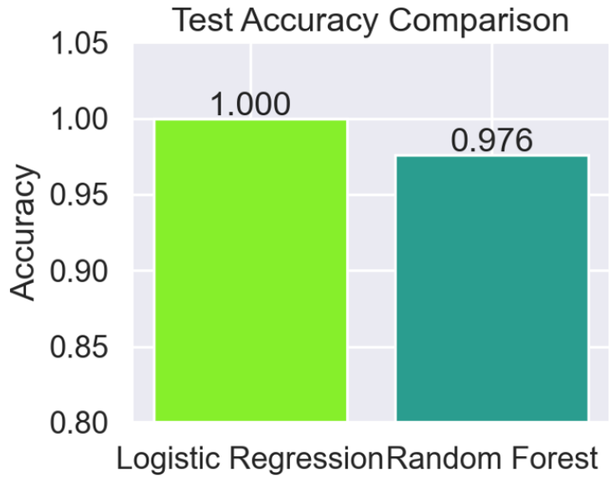
Stratified 5-fold cross validation

GridSearch hyperparameter tuning

Results & Evaluation

•I also contributed to model evaluation and interpretation. I generated evaluation metrics such as accuracy, precision, recall, and F1 score, and created confusion matrices to visualize model predictions and identify classification patterns.





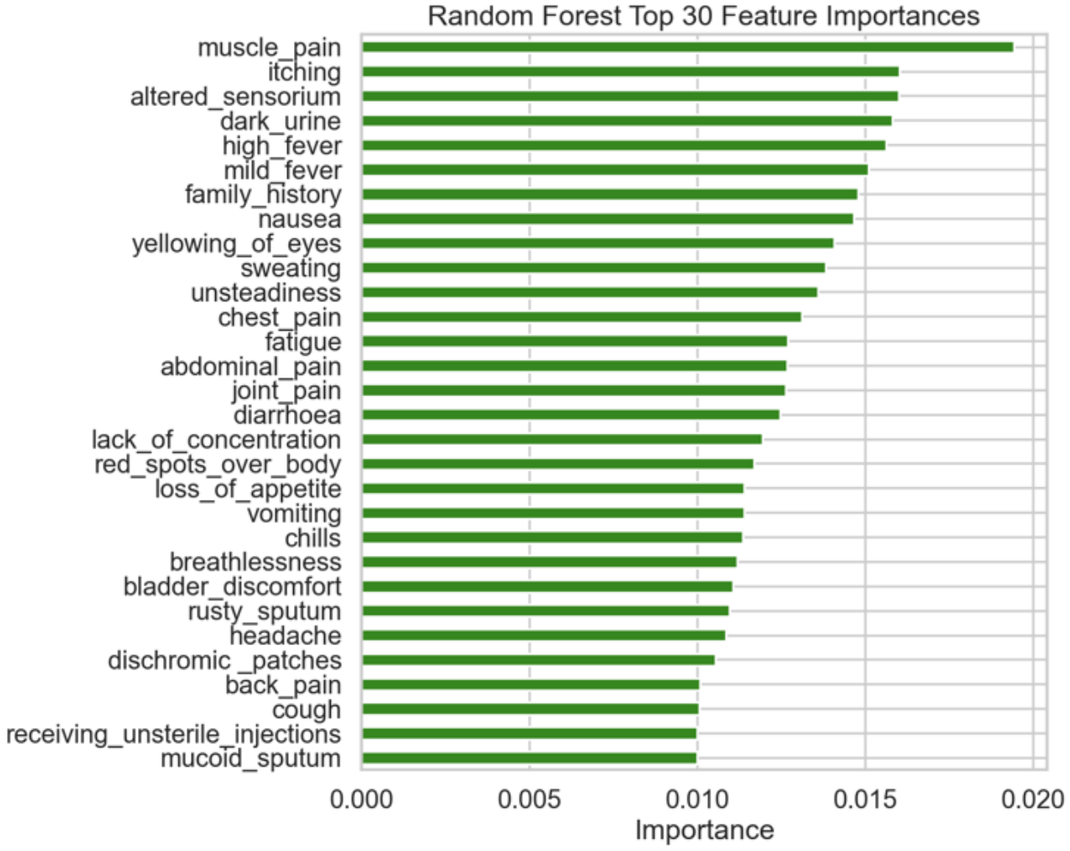
The high **accuracy** occurs because the dataset has very clear relationships between symptoms and diseases.

After training both models, evaluated them using: **Accuracy**, **Precision**, **Recall**, and **F1 Score**.

Results & Evaluation

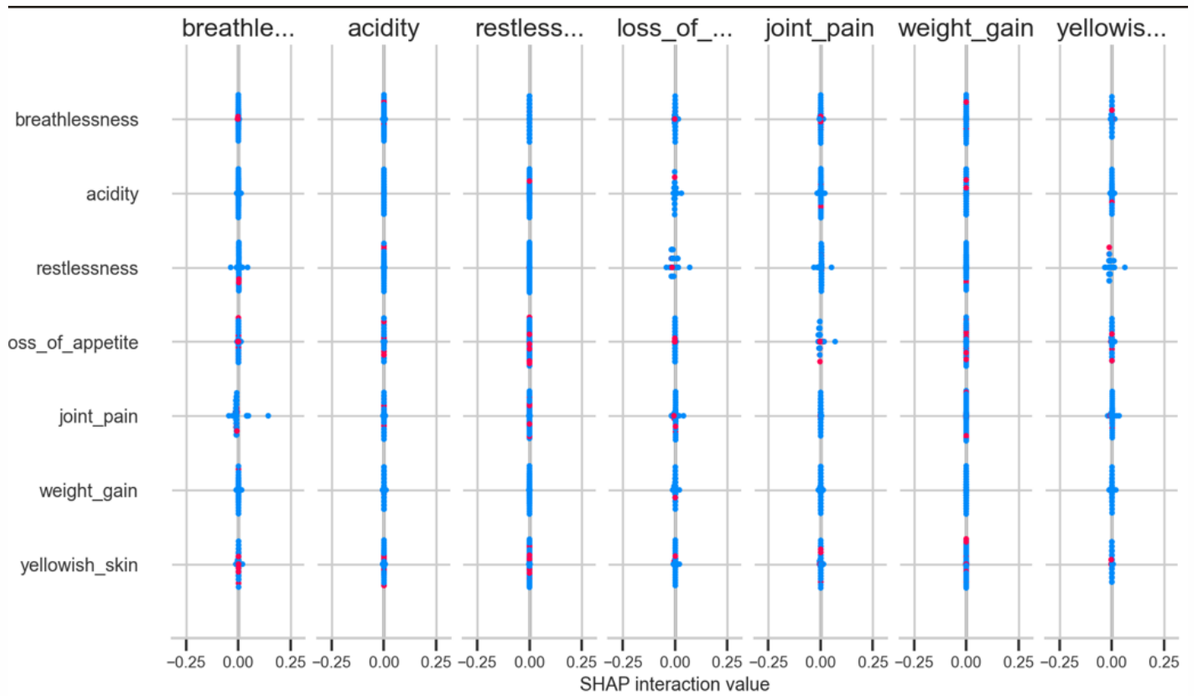
• In addition, I conducted model interpretability analysis by examining feature importance using Random Forest and applying SHAP explainability techniques to understand how individual symptoms influence the model's predictions.

These features contributed strongly to disease predictions.

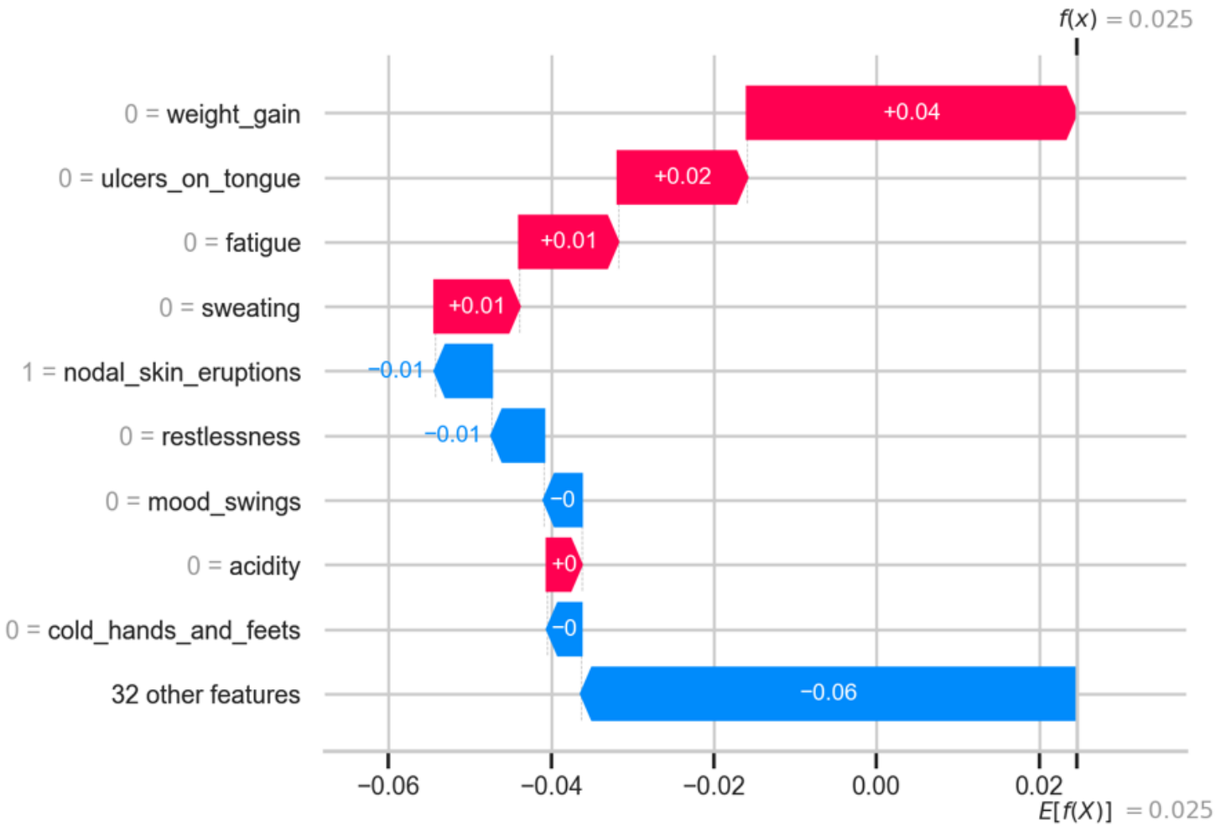


Results & Evaluation

Model Explain ability Using SHAP



SHAP Interaction Plot. shows how different symptoms interact with each other and how their combined effects influence the model's prediction.



SHAP waterfall plot explains how individual symptoms increase or decrease the model's prediction for a specific case.

Conclusion

In conclusion, this project demonstrates how **machine learning can be used to predict diseases based on symptoms**.

Model Performance

Both models performed well, with **Logistic Regression achieving the highest accuracy**. (100%)

Real-World Role

Although this system **cannot replace doctors**, it can serve as a **decision-support tool** for early disease detection.

Project Contribution Write-Up

- Beyond the modeling work, I played an active team member role within the team. I follow track project progress, aligned team members on project tasks and timelines, and assisted teammates in troubleshooting technical issues when needed.

- One challenge I faced was that the test dataset was unbalanced, which made model evaluation difficult. I addressed this by discussing the issue with my team members and technical instructor to better understand the problem and identify possible solutions.

- Given additional time, several improvements could further strengthen the project. First, testing the models on real clinical datasets would help evaluate performance under more realistic and noisy conditions. Second, extending the explainability analysis to a per-disease level could help identify the most discriminative symptoms for each condition.

- My strength in a team environment is collaborating with others, communicating ideas clearly, and contributing to problem solving.